

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Subject: 392132 Physics II

Date: 20 April 2019

Year: 2018

Section: 15-18

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 6 pages (including this page), 11 questions and they are worth 45 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can't be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. A baseball ($m = 0.2 \text{ kg}$) has an initial velocity of $\vec{v}_0 = -40 \text{ m/s}$ as it approaches a bat. The bat applies an average force F that is much larger than the weight of the ball, and the ball departs from the bat with a final velocity of $\vec{v}_f = 60 \text{ m/s}$.

a) Determine the impulse applied to the ball by the bat.

b) **Assume** that the time of contact is $\Delta t = 4 \times 10^{-3} \text{ s}$, find the average force exerted on the ball by the bat. (4 points)

2. During a storm, rain comes straight down with a velocity of $\vec{v}_0 = -15 \text{ m/s}$ and hits the roof of a car perpendicularly. The mass of rain per second that strikes the car roof is 0.050 kg/s . **Assume** that the rain comes to rest upon striking the car ($\vec{v}_f = 0 \text{ m/s}$), find the average force exerted by the rain on the roof. (3 points)

3. Starting from rest, two skaters push off against each other on smooth level ice, where friction is negligible. One is a woman ($m_1 = 50 \text{ kg}$), and one is a man ($m_2 = 100 \text{ kg}$) after that the woman moves away with a velocity of 2.0 m/s . Find

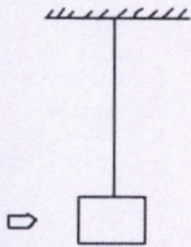
a) The recoil velocity of the man

b) ΔE_k

(4 points)

4. A two-stage rocket moves in space at a constant velocity of $4,000 \text{ m/s}$. The two stages are then separated by a small explosive charge placed between them. Immediately after the explosion the velocity of the $1,200 \text{ kg}$ upper stage is $5,000 \text{ m/s}$ in the same direction as before the explosion. What is the velocity (magnitude and direction) of the $1,800 \text{ kg}$ lower stage after the explosion? (3 points)

5. A ballistic pendulum can be used to measure the speed of a projectile, such as a bullet. The ballistic pendulum shown in Figure consists of a stationary 2.0 kg block of wood suspended by a wire of negligible mass. A 10 g bullet is fired into the block, and the block (with the bullet in it) swings to a maximum height of 0.20 m above the initial position. Find the speed with which the bullet is fired, assuming that air resistance is negligible. (4 points)

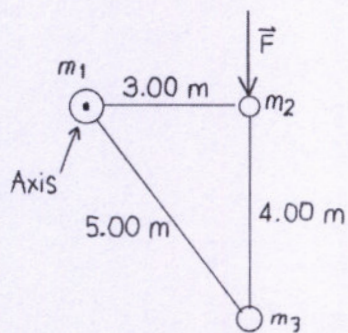


6. A 0.01 kg bullet is fired straight up at a falling wooden block that has a mass of 2 kg. the bullet has a speed of 800 m/s when it strikes the block. The block originally was dropped from rest from the top of a building and had been falling for a time t when the collision with the bullet occurred. As a result of the collision, the block (with the bullet in it) reverses direction, rises, and comes to a momentary halt at the top of the building. Find the time t . (6 points)

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7. From the figure, the system consists of the same three small balls connected by massless rods. The axis is perpendicular to the page. The masses of the balls are $m_1 = 9 \text{ kg}$, $m_2 = 6 \text{ kg}$, and $m_3 = 7 \text{ kg}$. The magnitude of the force is 500 N.

- Determine the moment of inertia about the given axis of rotation.
- Calculate the torque (magnitude and direction). (4 points)



8. The motor in an electric saw brings the circular blade from rest up to the rated angular velocity of 80 rev/s in 120π rev. One type of blade has a moment of inertia of $1.5 \times 10^{-3} \text{ kg} \cdot \text{m}^2$. What net torque (assumed constant) must the motor apply to the blade? (4 points)

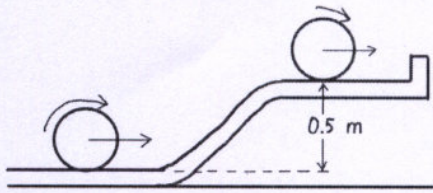
9. A 20.0-kg cylinder rolls without slipping on a rough surface. At an instant when its center of gravity has a speed of 10.0 m/s, determine

- a) the translational kinetic energy of its center of gravity,
- b) the rotational kinetic energy about its center of gravity, and
- c) its total kinetic energy.

(4 points)

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10. A bowling ball encounters a 0.5 m vertical rise on the way back to the ball rack, as the drawing illustrates. Ignore frictional losses and assume that the mass of the ball is distributed uniformly. The translational speed of the ball is 5 m/s at the bottom of the rise. Find the translational speed at the top. $\left(I_{\text{solid sphere}} = \frac{2}{5} mR^2 \right)$ (5 points)



11. A thin rod has a length of 0.8 m and rotates in a circle on a frictionless table top. The axis is perpendicular to the length of the rod at one of its end. The rod has an angular velocity of 0.6 rad/s and a moment of inertia of $1 \times 10^{-3} \text{ kg} \cdot \text{m}^2$. A bug standing on the axis decides to crawl out to the other end of the rod. When the bug (mass = $5 \times 10^{-3} \text{ kg}$) gets where it's going. What is the angular velocity of the rod?

(4 points)

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